ASSIGNMENT 3

SUBJECT CODE: CSU1658
SUBJECT NAME: Statistical Foundation of Data Sciences
TOTAL POINTS: 100
DEADLINE: DECEMBER 20TH 2025 4PM

K-Nearest Neighbors (KNN)

PROBLEM 1: Medical Diagnosis Classification

A hospital classifies patients based on glucose level (x_1) and BMI (x_2) as diabetic (+) or non-diabetic (-).

Training data:

Patient	Glucose	BMI	Class
A	85	22	$-$
B	95	25	$-$
C	140	32	$+$
D	160	35	$+$
E	120	28	$+$
F	90	24	$-$

New patient: $P = (110, 27)$.

(a) Compute Euclidean distances from P to all training patients. Normalize features first using $\tilde{x} = (x - \min)/(max - \min)$.

(b) Classify using $k = 3$ with majority voting. Show the complete decision process.

(c) Derive the decision boundary equation for 1-NN between two nearest neighbors algebraically.

PROBLEM 2: Curse of Dimensionality Analysis

Consider KNN in high-dimensional spaces.

(a) Prove that in d dimensions, the volume of a hypersphere of radius r is $V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$. Show that as $d \to \infty$, most volume concentrates near the surface.

(b) Given 1000 uniformly distributed points in $[0,1]^d$, derive the expected distance to the nearest neighbor as a function of d. Calculate for $d = 2, 10, 50$.

(c) Explain why KNN performance degrades in high dimensions. Propose two methods to mitigate this issue.

PROBLEM 3: Customer Segmentation with Manhattan Distance

E-commerce data: purchases per month (x_1) and average order value (x_2):

ID	Purchases	Avg Value	Segment
1	2	50	Low
2	8	120	High
3	5	80	Medium
4	12	200	High
5	3	60	Low
6	7	90	Medium

New customer: $C = (6, 100)$.

(a) Calculate Manhattan distances: $d_M(x, y) = |x_1 - y_1| + |x_2 - y_2|$. Why might Manhattan be better than Euclidean here?

(b) Classify C using weighted KNN with $k = 4$ and weights $w_i = e^{-d_i}$. Show all calculations.

(c) Implement leave-one-out cross-validation to find optimal $k \in \{1, 3, 5\}$. Which k minimizes error?

PROBLEM 4: KNN Regression for Temperature Prediction

Weather stations record altitude (m) and temperature (°C):

Station	Altitude	Temperature
S_1	100	28
S_2	500	22
S_3	1000	15
S_4	1500	10
S_5	2000	5

Predict temperature at altitude 1200m using KNN regression.

(a) Use $k = 3$ with inverse distance weighting: $\hat{y} = \frac{\sum_{i=1}^{k} w_i y_i}{\sum_{i=1}^{k} w_i}$ where $w_i = 1/d_i^2$. Calculate the predicted temperature.

(b) Derive the bias-variance decomposition for KNN regression. How does k affect bias and variance?

(c) Prove that as $k \to n$, KNN regression converges to the global mean. What does this imply about underfitting?

PROBLEM 5: KD-Tree Construction and Complexity

Dataset in 2D: $\{(2,3),(5,4),(9,6),(4,7),(8,1),(7,2)\}$.

(a) Construct a KD-tree by recursively splitting on alternating dimensions. Draw the tree structure and partitioning regions.

(b) Search for the nearest neighbor of query point $(6,5)$ using the KD-tree. Show the pruning decisions and nodes visited.

(c) Prove that KD-tree query time is $O(\log n)$ on average for balanced trees. When does it degrade to $O(n)$?

K-Means Clustering

PROBLEM 6: Image Color Quantization

Image has 9 pixels with RGB values (simplified):

Pixel	R	G	B
1	255	0	0
2	250	10	5
3	245	5	10
4	0	255	0
5	5	250	10
6	10	245	5
7	0	0	255
8	5	10	250
9	10	5	245

(a) Run K-means with $k = 3$ initialized at pixels 1, 4, 7. Show two complete iterations (assignment + update).

(b) Calculate final WCSS and explain how this reduces image file size.

(c) Derive the relationship between K-means and vector quantization in signal processing.

PROBLEM 7: K-Means++ Initialization

Consider points: $\{1, 4, 8, 12, 15, 18, 22\}$ in 1D, $k = 3$.

(a) Implement K-means++ initialization: choose first centroid randomly, then select subsequent centroids with probability proportional to $D(x)^2$ (squared distance to nearest centroid). Show the probability distribution at each step.

(b) Compare final WCSS from K-means++ vs random initialization $\{\mu_1 = 1, \mu_2 = 4, \mu_3 = 8\}$.

(c) Prove that K-means++ gives an $O(\log k)$-approximation to optimal clustering in expectation.

PROBLEM 8: Soft K-Means (Gaussian Mixture Model)

Dataset: $\{0.5, 1.5, 2.0, 8.5, 9.0, 9.5\}$, $k = 2$.

(a) Derive the soft K-means (EM for GMM) update equations:

$$r_{ik} = \frac{\exp(-\beta\|x_i - \mu_k\|^2)}{\sum_j \exp(-\beta\|x_i - \mu_j\|^2)}$$

where $\beta = 1/(2\sigma^2)$.

(b) Run one EM iteration with $\mu_1 = 1, \mu_2 = 9, \sigma^2 = 1$. Calculate responsibilities r_{ik} and update $\mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}$.

(c) Compare hard vs soft clustering. When is soft clustering preferred?

PROBLEM 9: Spectral Clustering Foundation

Similarity graph with 5 nodes and adjacency matrix:

$$W = \begin{pmatrix} 0 & 8 & 2 & 0 & 0 \\ 8 & 0 & 7 & 0 & 0 \\ 2 & 7 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 9 & 0 \end{pmatrix}$$

(a) Compute the degree matrix D and Laplacian $L = D - W$. Find eigenvalues and eigenvectors of L.

(b) Use the Fiedler vector (second smallest eigenvalue's eigenvector) to partition the graph into 2 clusters.

(c) Explain why spectral clustering can find non-convex clusters while K-means cannot. Give a geometric example.

PROBLEM 10: Hierarchical vs K-Means

Gene expression levels (simplified): $\{2.1, 2.5, 2.3, 7.8, 8.2, 7.5, 15.1, 15.5\}$.

(a) Perform agglomerative hierarchical clustering with complete linkage. Draw the dendrogram.

(b) Cut dendrogram to get 3 clusters. Run K-means with $k = 3$ (random init). Compare cluster assignments.

(c) Prove that hierarchical clustering has time complexity $O(n^3)$ for complete linkage while K-means is $O(ikn)$. Discuss trade-offs.

Principal Component Analysis (PCA)

PROBLEM 11: Stock Portfolio Dimensionality Reduction

Returns of 3 stocks over 4 days (percentages):

$$X = \begin{pmatrix} 2 & 3 & 2.5 \\ 1 & 2 & 1.5 \\ -1 & -2 & -1.5 \\ -2 & -3 & -2.5 \end{pmatrix}$$

(a) Center the data and compute covariance matrix $\Sigma = \frac{1}{n-1} X^T X$. Interpret Σ_{12}.

(b) Find eigenvalues and eigenvectors. Calculate the first principal component direction and interpret it financially.

(c) Project data onto PC1. What percentage of portfolio variance is explained by this single factor?

PROBLEM 12: PCA via SVD for Tall Matrix

Data matrix (8 observations, 2 features):

$$X = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 5 \\ 4 & 7 \\ 5 & 9 \\ 6 & 11 \\ 7 & 12 \\ 8 & 14 \end{pmatrix}$$

(assume centered)

(a) Compute SVD: $X = U\Sigma V^T$. Show that principal components are columns of V.

(b) Derive the relationship: eigenvalues of $X^T X$ equal σ_i^2 (squared singular values).

(c) Calculate reconstruction error when approximating X with rank-1 matrix: $\|X - X_1\|_F^2$. Verify it equals σ_2^2.

PROBLEM 13: Kernel PCA for Nonlinear Data

1D data: $X = \{-3, -1, 0, 1, 3\}$ with nonlinear relationship.

(a) Map to 2D using $\phi(x) = (x, x^2)$. Compute the mapped data matrix and its covariance.

(b) Derive the kernel matrix $K_{ij} = \phi(x_i)^T \phi(x_j) = x_i x_j + x_i^2 x_j^2$ without explicit feature mapping.

(c) Perform kernel PCA by eigendecomposition of centered kernel matrix. Compare with standard PCA on original data.

PROBLEM 14: PCA for Face Recognition (Eigenfaces)

Grayscale 2×2 face images (vectorized to 4D):

$$X = \begin{pmatrix} 200 & 180 & 190 & 185 \\ 180 & 200 & 185 & 195 \\ 80 & 90 & 85 & 88 \\ 20 & 30 & 25 & 28 \end{pmatrix}$$

(rows are pixels, columns are different faces)

(a) Compute mean face $\bar{x}$ and center data. Find eigenfaces (principal components).

(b) Represent a new face $f_{\text{new}} = [195, 188, 82, 22]^T$ using top 2 eigenfaces.

(c) Derive the reconstruction and compare with original. Calculate mean squared error.

PROBLEM 15: Probabilistic PCA

Observed data: $x \in \mathbb{R}^d$ generated from latent variable $z \in \mathbb{R}^q$ where $q < d$.
Model: $x = Wz + \mu + \epsilon$ where $z \sim \mathcal{N}(0, I)$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

(a) Derive the marginal distribution: $x \sim \mathcal{N}(\mu, WW^T + \sigma^2 I)$.

(b) Show that maximum likelihood estimate of W gives: $W_{ML} = V_q(\Lambda_q - \sigma^2 I)^{1/2} R$ where V_q contains top q eigenvectors and R is arbitrary rotation.

(c) Prove that as $\sigma^2 \to 0$, probabilistic PCA reduces to standard PCA. What's the advantage of the probabilistic formulation?

Gaussian Distribution

PROBLEM 16: Quality Control with Normal Distribution

Manufacturing process produces bolts with diameter $\sim \mathcal{N}(10, 0.04)$ mm. Specifications: 9.7–10.3 mm.

(a) Calculate probability a random bolt meets specifications. Use error function: $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.

(b) If 10,000 bolts are produced daily, how many defective bolts are expected?

(c) Derive the process capability index $C_p = \frac{USL-LSL}{6\sigma}$. If $C_p < 1.33$, what σ is needed?

PROBLEM 17: Multivariate Gaussian Conditional Distribution

Bivariate distribution: height X_1 and weight X_2 with

$$\boldsymbol{\mu} = \begin{pmatrix} 170 \\ 70 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 64 & 32 \\ 32 & 25 \end{pmatrix}$$

(a) Derive the conditional distribution $X_2|X_1 = 180$ using the formula:

$$\mu_{2|1} = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1), \quad \Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

(b) Calculate expected weight for height = 180 cm and the conditional variance.

(c) Prove the conditional distribution is also Gaussian. Why is this property important in Bayesian inference?

PROBLEM 18: Chi-Square Goodness-of-Fit Test

Dice rolled 600 times with observed frequencies:

Face	1	2	3	4	5	6
Observed	92	107	98	95	103	105

(a) Test H_0: dice is fair using $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$. Calculate test statistic.

(b) Find critical value for $\alpha = 0.05$ with appropriate df. Make decision about fairness.

(c) Derive the asymptotic distribution of χ^2 statistic using central limit theorem. Why does it follow chi-square distribution?

PROBLEM 19: Gaussian Mixture Model Classification

Two classes with Gaussian distributions:

- Class 1: $\mathcal{N}(2, 1)$, prior $\pi_1 = 0.6$
- Class 2: $\mathcal{N}(5, 1.5)$, prior $\pi_2 = 0.4$

(a) Derive Bayes optimal decision boundary by solving: $\pi_1 p(x|C_1) = \pi_2 p(x|C_2)$.

(b) Classify observation $x = 3.5$ using posterior probabilities: $P(C_i|x) = \frac{\pi_i p(x|C_i)}{\sum_j \pi_j p(x|C_j)}$.

(c) Calculate expected misclassification rate for this decision rule.

PROBLEM 20: Q-Q Plot for Normality Testing

Sample data (sorted): $\{12.3, 14.1, 15.8, 16.2, 17.5, 18.9, 20.1, 22.4\}$.

(a) Calculate theoretical quantiles from $\mathcal{N}(0, 1)$ for plotting positions $p_i = \frac{i-0.5}{n}$. Use $\Phi^{-1}(p_i)$.

(b) Standardize data: $z_i = \frac{x_i - \bar{x}}{s}$. Plot $(z_{(i)}, q_i)$ and assess linearity.

(c) Perform Shapiro-Wilk test using: $W = \frac{(\sum a_i x_{(i)})^2}{\sum (x_i - \bar{x})^2}$. Interpret result ($W$ close to 1 indicates normality).

Ridge Regression

PROBLEM 21: Housing Price Prediction with Multicollinearity

Predicting house price from size (x_1) and rooms (x_2) (size and rooms are highly correlated):

$$X = \begin{pmatrix} 1 & 1000 & 2 \\ 1 & 1500 & 3 \\ 1 & 2000 & 4 \\ 1 & 2500 & 5 \end{pmatrix}, \quad y = \begin{pmatrix} 200 \\ 270 \\ 340 \\ 410 \end{pmatrix}$$

(prices in thousands)

(a) Compute $X^T X$ and calculate its condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. What does high κ indicate?

(b) Derive and compute ridge solution with $\lambda = 100$: $\hat{\beta}^{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$.

(c) Compare with OLS. Show how ridge shrinks coefficients and reduces variance at cost of bias.

PROBLEM 22: Cross-Validation for Lambda Selection

Small dataset: $X = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 8 \end{pmatrix}$, $y = \begin{pmatrix} 3 \\ 7 \\ 9 \\ 13 \end{pmatrix}$.

(a) Implement leave-one-out cross-validation for $\lambda \in \{0, 1, 5, 10\}$. Calculate CV error: $\text{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_{-i}^{(\lambda)})^2$.

(b) Derive the LOOCV shortcut formula: $CV(\lambda) = \frac{1}{n} \sum \left(\frac{y_i - \hat{y}_i}{1 - H_{ii}} \right)^2$ where $H = X(X^T X + \lambda I)^{-1} X^T$.

(c) Plot CV error vs λ. Which λ is optimal? Explain bias-variance trade-off.

PROBLEM 23: Ridge Regression as Constrained Optimization

Consider equivalent formulations of ridge regression.

(a) Show that ridge regression $\min_\beta \|y - X\beta\|^2 + \lambda \|\beta\|^2$ is equivalent to constrained problem: $\min_\beta \|y - X\beta\|^2$ subject to $\|\beta\|^2 \le t$ for some $t(\lambda)$.

(b) Derive the Lagrangian and KKT conditions. Find the relationship between λ and constraint t.

(c) Geometrically illustrate ridge solution as intersection of constraint ball $\|\beta\| \le \sqrt{t}$ and OLS error ellipsoids. Why does ridge shrink towards origin?

PROBLEM 24: Generalized Cross-Validation (GCV)

For design matrix X and penalty λ.

(a) Derive the GCV criterion: $\text{GCV}(\lambda) = \frac{n\|y - X\hat{\beta}^{\lambda}\|^2}{(n - \text{df}(\lambda))^2}$ where $\text{df}(\lambda) = \text{Tr}[X(X^T X + \lambda I)^{-1} X^T]$.

(b) For dataset: $X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$, $y = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$, compute $\text{GCV}(\lambda)$ for $\lambda = 0.5$.

(c) Explain why GCV approximates leave-one-out CV but is computationally cheaper.

PROBLEM 25: Tikhonov Regularization (Generalized Ridge)

Generalized ridge with penalty matrix Γ: $\min_\beta \|y - X\beta\|^2 + \lambda\|\Gamma\beta\|^2$.

(a) Derive solution: $\hat{\beta} = (X^T X + \lambda \Gamma^T \Gamma)^{-1} X^T y$. For $\Gamma = D$ (difference matrix), what constraints are imposed?

(b) Consider trend filtering with $D = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ (first differences). Solve for data: $X = I_3$, $y = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$, $\lambda = 1$.

(c) Relate Tikhonov regularization to prior $\beta \sim \mathcal{N}(0, (\lambda \Gamma^T \Gamma)^{-1})$. What does $\Gamma = D$ encode?

Controlling Standard Deviation

PROBLEM 26: Pharmaceutical Tablet Weight Variance

Tablet weight regulation requires $\sigma \leq 2$ mg. Sample of 25 tablets has $s = 2.3$ mg.

(a) Test $H_0 : \sigma^2 = 4$ vs $H_a : \sigma^2 > 4$ at $\alpha = 0.05$ using $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$. Critical value: $\chi^2_{0.05,24} = 36.42$.

(b) Calculate the p-value: $P(\chi^2_{24} > \chi^2_{\text{obs}})$. Make regulatory decision.

(c) Derive power of test: $1-\beta = P(\text{reject } H_0|\sigma^2 = 6)$. How many samples needed for 80% power?

PROBLEM 27: Comparing Process Variabilities

Two assembly lines produce resistors. Line A: $n_1 = 16, s_1^2 = 0.49$. Line B: $n_2 = 21, s_2^2 = 0.25$.

(a) Test equality of variances: $H_0 : \sigma_1^2 = \sigma_2^2$ using $F = s_1^2/s_2^2 \sim F_{15,20}$. Critical values: $F_{0.025,15,20} = 2.76$, $F_{0.975,15,20} = 0.39$.

(b) Construct 95% CI for variance ratio: $\left[\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{\alpha/2}}, \frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{1-\alpha/2}}\right]$.

(c) If variances differ significantly, which statistical tests become invalid? Suggest robust alternatives.

PROBLEM 28: Bartlett's Test for Homogeneity of Variances

Three groups with sample sizes $n = (10, 12, 15)$ and variances $s^2 = (4.2, 3.8, 5.1)$.

(a) Calculate pooled variance: $s_p^2 = \frac{\sum(n_i-1)s_i^2}{\sum(n_i-1)}$.

(b) Compute Bartlett's statistic: $B = \frac{(N-k)\ln(s_p^2) - \sum(n_i-1)\ln(s_i^2)}{1+\frac{1}{3(k-1)}[\sum\frac{1}{n_i-1} - \frac{1}{N-k}]}$ where $N = \sum n_i$, $k = 3$.

(c) Under H_0: equal variances, $B \sim \chi^2_{k-1}$. Test at $\alpha = 0.05$ with $\chi^2_{0.05,2} = 5.99$.

PROBLEM 29: Brown-Forsythe Test (Robust to Non-Normality)

Data from 3 groups (deviations from median):

- Group 1: $|x_i - \text{med}_1| = \{2.1, 1.8, 2.3, 1.9\}$
- Group 2: $|x_i - \text{med}_2| = \{3.2, 2.9, 3.5, 3.1\}$
- Group 3: $|x_i - \text{med}_3| = \{1.5, 1.7, 1.4, 1.6\}$

(a) Perform ANOVA on absolute deviations to test variance equality. Calculate $F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$.

(b) Why is Brown-Forsythe preferred over Bartlett's for non-normal data? Explain robustness.

(c) If H_0 rejected, perform post-hoc pairwise comparisons using Bonferroni correction. Which groups differ?

PROBLEM 30: Cochran's C Test for Outlier Variance

Experiment conducted 6 times with sample variances: $s^2 = \{4.1, 3.9, 4.2, 3.8, 4.0, 12.5\}$ (all $n_i = 5$).

(a) Calculate Cochran's statistic: $C = \frac{\max(s_i^2)}{\sum s_i^2} = \frac{12.5}{32.5}$.

(b) Compare with critical value $C_{0.05}(6, 4) = 0.616$ (6 groups, $df = 4$ each). Is largest variance an outlier?

(c) If outlier detected, investigate: run Grubbs' test on original data from that group. Derive decision rule.

Markov Chains

PROBLEM 31: Brand Loyalty Model

Customers switch between brands A, B, C monthly:

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.6 & 0.1 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Current market shares: $\pi^{(0)} = (0.4, 0.35, 0.25)$.

(a) Calculate market shares after 3 months: $\pi^{(3)} = \pi^{(0)} P^3$. Show intermediate steps.

(b) Find equilibrium market shares by solving $\pi P = \pi$, $\sum \pi_i = 1$. Use substitution or eigenvalue method.

(c) Derive mean return time to brand A: $\tau_A = 1/\pi_A$. Interpret in business context.

PROBLEM 32: Absorbing Markov Chain - Credit Rating

Bond ratings: AAA, AA, A, D (default, absorbing). Transition matrix:

$$P = \begin{pmatrix} 0.90 & 0.08 & 0.02 & 0 \\ 0.05 & 0.85 & 0.08 & 0.02 \\ 0.02 & 0.10 & 0.80 & 0.08 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) Identify transient and absorbing states. Write canonical form: $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$.

(b) Calculate fundamental matrix $N = (I - Q)^{-1}$. Interpret N_{ij} as expected visits to state j starting from i.

(c) Find absorption probabilities: $B = NR$. What's probability an AA bond eventually defaults?

PROBLEM 33: PageRank Algorithm

Web graph with 4 pages and link structure:

$$L = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

(a) Form Google matrix: $G = \alpha L + (1 - \alpha)\frac{\mathbf{1}\mathbf{1}^T}{n}$ with $\alpha = 0.85$. Why is teleportation term needed?

(b) Compute PageRank by power iteration: $\pi^{(k+1)} = G^T \pi^{(k)}$ starting from $\pi^{(0)} = (0.25, 0.25, 0.25, 0.25)$. Iterate until $\|\pi^{(k+1)} - \pi^{(k)}\| < 10^{-3}$.

(c) Prove convergence using Perron-Frobenius theorem. What conditions ensure unique stationary distribution?

PROBLEM 34: Birth-Death Process for Queuing

Server with arrival rate $\lambda = 3$ customers/hr, service rate $\mu = 4$ customers/hr.

Generator matrix for states $n = 0, 1, 2, 3$ (max capacity):

$$Q = \begin{pmatrix} -3 & 3 & 0 & 0 \\ 4 & -7 & 3 & 0 \\ 0 & 4 & -7 & 3 \\ 0 & 0 & 4 & -4 \end{pmatrix}$$

(a) Verify detailed balance: $\pi_i q_{ij} = \pi_j q_{ji}$. Derive stationary distribution $\pi_n = \pi_0(\lambda/\mu)^n$.

(b) Calculate average queue length: $L = \sum n\pi_n$ and expected wait time using Little's law: $W = L/\lambda$.

(c) Find probability of system being full (customer rejected). How does this change if μ increases to 5?

PROBLEM 35: Hidden Markov Model Basics

2-state HMM: Sunny (S) and Rainy (R) with transitions and emissions.

Transition: $A = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$. Emission: $B = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}$ (Happy/Grumpy).

Observe: Happy, Happy, Grumpy.

(a) Calculate $P(\text{observations}|\text{states} = SSR)$ using $P(O|\lambda) = \prod P(o_t|s_t) \cdot P(s_t|s_{t-1})$ with initial $\pi = (0.6, 0.4)$.

(b) Implement forward algorithm to compute $P(\text{observations})$: $\alpha_t(i) = P(o_1, \ldots, o_t, s_t = i)$.

(c) Use Viterbi algorithm to find most likely state sequence. Show trellis diagram.

Monte Carlo Methods

PROBLEM 36: Buffon's Needle for Estimating Pi

Needle of length $L = 1$ dropped on floor with parallel lines distance $D = 2$ apart.

(a) Derive probability needle crosses a line: $P = \frac{2L}{\pi D}$. (Hint: parameterize by angle θ and distance x from nearest line).

(b) Simulate 10,000 drops. Estimate π using $\hat{\pi} = \frac{2L \cdot n_{\text{total}}}{D \cdot n_{\text{cross}}}$. Calculate standard error.

(c) Compare efficiency with circle-in-square method. Which has lower variance?

PROBLEM 37: Stratified Sampling for Variance Reduction

Estimate $I = \int_0^4 x^3 dx$ (true value = 64).

(a) Standard MC with $n = 100$ uniform samples on $[0, 4]$: $\hat{I}_{\text{MC}} = 4 \cdot \frac{1}{n} \sum f(U_i)$. Calculate variance.

(b) Stratified sampling: divide $[0, 4]$ into 4 equal strata. Sample 25 points per stratum. Calculate $\hat{I}_{\text{strat}} = \sum w_k \bar{f}_k$ where $w_k = 0.25$.

(c) Prove variance reduction: $\text{Var}(\hat{I}_{\text{strat}}) = \sum w_k^2 \sigma_k^2 / n_k < \text{Var}(\hat{I}_{\text{MC}})$. Verify numerically.

PROBLEM 38: Control Variates Method

Estimate $\theta = E[e^X]$ where $X \sim \text{Uniform}(0, 1)$.
Control variate: $Y = X$ with known $E[Y] = 0.5$.

(a) Simulate $n = 1000$ samples. Estimate $\hat{\theta}_0 = \frac{1}{n} \sum e^{X_i}$ and calculate sample variance.

(b) Use control variate estimator: $\hat{\theta}_c = \hat{\theta}_0 - c(\bar{Y} - 0.5)$ where $c = \text{Cov}(e^X, X)/\text{Var}(X)$. Estimate c from sample covariance.

(c) Show variance reduction: $\text{Var}(\hat{\theta}_c) = \text{Var}(\hat{\theta}_0)(1 - \rho^2)$ where $\rho = \text{Cor}(e^X, X)$. Calculate efficiency gain.

PROBLEM 39: Gibbs Sampling for Bivariate Normal

Target: $(X, Y) \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right)$.
Conditionals: $X|Y \sim \mathcal{N}(0.8Y, 0.36)$, $Y|X \sim \mathcal{N}(0.8X, 0.36)$.

(a) Implement Gibbs sampler for 5000 iterations starting at $(X_0, Y_0) = (0, 0)$. Plot trace of X_t.

(b) Calculate autocorrelation: $\rho_k = \text{Cor}(X_t, X_{t+k})$ for lags $k = 1, \ldots, 50$. Estimate effective sample size: $n_{\text{eff}} = \frac{n}{1+2\sum \rho_k}$.

(c) Compare marginal histograms with true $\mathcal{N}(0, 1)$. Perform Kolmogorov-Smirnov test for goodness of fit.

PROBLEM 40: Rejection Sampling from Truncated Distribution

Target: truncated normal $X \sim \mathcal{N}(0, 1)$ on $[2, \infty)$.

Proposal: exponential $g(x) = e^{-(x-2)}$ for $x \geq 2$.

(a) Find constant M such that $f(x) \leq Mg(x)$ for all $x \geq 2$ where $f(x) = \frac{\phi(x)}{\Phi(-2)}$.

(b) Implement rejection algorithm:

   (a) Generate $Y \sim g$, $U \sim \text{Uniform}(0, 1)$

   (b) Accept if $U \leq f(Y)/(Mg(Y))$

   Run until 1000 samples accepted. Calculate acceptance rate.

(c) Derive expected number of proposals per acceptance: $1/P_{\text{accept}} = M$. Discuss efficiency vs direct sampling.

Deterministic Models

PROBLEM 41: Pharmacokinetics - Drug Concentration Model

One-compartment model: $\frac{dC}{dt} = -kC$ where $C(t)$ is drug concentration, $k = 0.2$ hr^{-1}.
Single dose: $C(0) = 100$ mg/L. Repeated dosing: $D = 50$ mg/L every 4 hours.

(a) Solve ODE for single dose: $C(t) = C_0 e^{-kt}$. Calculate half-life: $t_{1/2} = \ln(2)/k$.

(b) For repeated dosing, derive steady-state concentration: $C_{ss} = \frac{D}{1-e^{-kT}}$ where $T = 4$ hr is dosing interval. Calculate C_{ss}.

(c) Find therapeutic window $[C_{\min}, C_{\max}] = [20, 80]$. Adjust D and T to maintain $C(t)$ in this range.

PROBLEM 42: Chemical Reaction Kinetics

Second-order reaction: $A + B \to C$ with rate law $\frac{d[A]}{dt} = -k[A][B]$.
Initial: $[A]_0 = [B]_0 = 1.0$ M, $k = 0.5$ M^{-1}min^{-1}.

(a) For equal initial concentrations, derive: $\frac{1}{[A]} - \frac{1}{[A]_0} = kt$. Solve for $[A](t)$.

(b) Calculate time for 75% conversion ($[A] = 0.25$ M). At what time does $[A] = 0.1[A]_0$?

(c) If $[A]_0 \neq [B]_0$, derive: $\ln \frac{[A][B]_0}{[B][A]_0} = ([B]_0 - [A]_0)kt$. Solve for $[A]_0 = 1.0$, $[B]_0 = 2.0$.

PROBLEM 43: Zombie Apocalypse Model (SZR)

Population dynamics:

$$\begin{cases} \frac{dS}{dt} = -\beta SZ \\ \frac{dZ}{dt} = \beta SZ + \zeta R - \alpha SZ \\ \frac{dR}{dt} = \alpha SZ - \zeta R \end{cases}$$

Parameters: $\beta = 0.0001$ (infection), $\alpha = 0.0001$ (kill), $\zeta = 0.01$ (resurrection). Initial: $S(0) = 1000, Z(0) = 10, R(0) = 0$.

(a) Verify conservation when $\zeta = 0$: $S + Z + R = N$. What happens with $\zeta > 0$?

(b) Simulate for 100 days using Euler method with $\Delta t = 0.1$. Plot $S(t), Z(t), R(t)$.

(c) Find zombie-free equilibrium. Derive stability condition (zombie extinction threshold).

PROBLEM 44: Van der Pol Oscillator (Nonlinear Dynamics)

$$\frac{d^2x}{dt^2} - \mu(1-x^2)\frac{dx}{dt} + x = 0$$

Rewrite as system: $\frac{dx}{dt} = y$, $\frac{dy}{dt} = \mu(1-x^2)y - x$.

(a) Find fixed points. For $\mu = 1$, analyze stability using Jacobian at origin.

(b) Simulate with initial conditions $(x_0, y_0) = (0.1, 0)$ for $\mu \in \{0.1, 1, 5\}$ over $t \in [0, 50]$. Plot phase portraits.

(c) Explain limit cycle behavior. How does period depend on μ? (numerical approximation acceptable)

PROBLEM 45: Optimal Control - Vaccination Strategy

SIR model with vaccination: control $u(t) \in [0, 0.9]$ (fraction vaccinated daily).

$$\begin{cases} \frac{dS}{dt} = -\beta SI - u(t)S \\ \frac{dI}{dt} = \beta SI - \gamma I \\ \frac{dR}{dt} = \gamma I + u(t)S \end{cases}$$

Objective: minimize $J = \int_0^T [\alpha I(t) + \beta u(t)^2] dt$ (balance infection cost vs vaccination cost).

Parameters: $\beta = 0.0005, \gamma = 0.1, \alpha = 1, \beta = 10, T = 100$.

(a) Write Hamiltonian: $H = \alpha I + \beta u^2 + \lambda_S(-\beta SI - uS) + \lambda_I(\beta SI - \gamma I) + \lambda_R(\gamma I + uS)$.

(b) Derive optimal control from $\frac{\partial H}{\partial u} = 0$: $u^* = \max\{0, \min\{0.9, -\lambda_S S/(2\beta)\}\}$.

(c) Implement forward-backward sweep method to solve. Compare total cost with constant vaccination $u = 0.3$.